

Palau International Ship Registry



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MARINE CIRCULAR 183.1

To: RECOGNIZED ORGANIZATIONS, SHIPOWNERS, MANAGERS, OPERATORS, MASTERS, DEPUTY REGISTRARS, AND FLAG STATE INSPECTORS.

Subject: AMENDMENTS TO THE INTERNATIONAL CODE FOR FIRE SAFETY SYSTEM (FSS CODE)

1. Reference
 - 1.1 Resolution MSC.403(96) *Amendments to the International Code for Fire Safety Systems (FSS Cod)* Adopted on May 19, 2016
 - 1.2 Resolution MSC.410(97) *Amendments to the International Code for Fire Safety Systems (FSS Code)* Adopted on November 25, 2016
 - 1.3 Fire Safety System (FSS Code), as amended
2. Purpose
 - 2.1 The purpose of this Marine Circular is to inform shipowners, managers, operators, and Recognized Organizations about the content of Resolutions MSC.403 (96) and the Resolutions MSC.410(97) which are in force since January 1st 2020.
3. Applicability
 - 3.1 This Marine Circular applies to Palau registered vessels that to which the FSS Code applies.
4. General Requirements
 - 4.1 Palau Ship Registry Administrator requires that all Palau registered vessels of 500 GT or more, shall maintain a FSS Code and amendments, thereof, onboard.
 - 4.2 Recognized Organizations that are approved by the Palau Ship Registry Administrator, are required to take actions for the proper implementation of the above-mentioned amendments.
 - 4.3 Both Resolution MSC.403(96) and Resolution MSC.410(97) are attached to this Marine Circular.
5. Resolution MSC.403(96)
 - 5.1 The Resolution adopts the amendments to the FSS Code where a new Chapter 17 is added and details the requirements for foam firefighting appliances for the protection of helicopter facilities.



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6. Resolution MSC.410(97)

6.1 The resolution adopts amendments to FSS Code related to the arrangement of means of escape for passenger ships. The amendments will enter into force on 1 January 2020.

7. Contact

9.1 For further questions, please do not hesitate to contact the Technical Department of PISR at technical@palaureg.com

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RESOLUTION MSC.410(97)

(adopted on 25 November 2016)

AMENDMENTS TO THE INTERNATIONAL CODE FOR FIRE SAFETY SYSTEMS (FSS CODE)

THE MARITIME SAFETY COMMITTEE,

RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,

NOTING resolution MSC.98(73), by which it adopted the International Code for Fire Safety Systems ("the FSS Code"), which has become mandatory under chapter II-2 of the International Convention for the Safety of Life at Sea, 1974 ("the Convention"),

NOTING ALSO article VIII(b) and regulation II 2/3.22 of the Convention concerning the procedure for amending the FSS Code,

HAVING CONSIDERED, at its ninety-seventh session, amendments to the FSS Code proposed and circulated in accordance with article VIII(b)(i) of the Convention,

- 1) ADOPTS, in accordance with article VIII(b)(iv) of the Convention, amendments to the FSS Code the text of which is set out in the annex to the present resolution.
- 2) DETERMINES, in accordance with article VIII(b)(vi)(2)(bb) of the Convention, that the said amendments shall be deemed to have been accepted on 1 July 2019 unless, prior to that date, more than one third of the Contracting Governments to the Convention or Contracting Governments the combined merchant fleets of which constitute not less than 50% of the gross tonnage of the world's merchant fleet, have notified the Secretary-General of their objections to the amendments.
- 3) INVITES Contracting Governments to the Convention to note that, in accordance with article VIII(b)(vii)(2) of the Convention, the amendments shall enter into force on 1 January 2020 upon their acceptance in accordance with paragraph 2 above.
- 4) REQUESTS the Secretary-General, for the purposes of article VIII(b)(v) of the Convention, to transmit certified copies of the present resolution and the text of the amendments contained in the annex to all Contracting Governments to the Convention.
- 5) REQUESTS ALSO the Secretary-General to transmit copies of this resolution and its annex to Members of the Organization which are not Contracting Governments to the Convention.



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ANNEX

AMENDMENTS TO THE INTERNATIONAL CODE FOR FIRE SAFETY SYSTEMS (FSS CODE)

CHAPTER 13

ARRANGEMENT OF MEANS OF ESCAPE

In paragraph 2.1.2.2.2.1, the text of case 2 is replaced with the following: "Case 2: Passengers in public spaces occupied to 3/4 of maximum capacity, 1/3 of the crew distributed in public spaces; service spaces occupied by 1/3 of the crew: and crew accommodation occupied by 1/3 of the crew."



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RESOLUTION MSC.403(96) (adopted on 19 May 2016)

AMENDMENTS TO THE INTERNATIONAL CODE FOR FIRE SAFETY SYSTEMS (FFS CODE)

THE MARITIME SAFETY COMMITTEE,

RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,

NOTING resolution MSC.98(73), by which it adopted the International Code for Fire Safety Systems ("the FSS Code"), which has become mandatory under chapter II-2 of the International Convention for the Safety of Life at Sea, 1974 ("the Convention"),

NOTING ALSO article VIII(b) and regulation II-2/3.22 of the Convention concerning the procedure for amending the FSS Code,

HAVING CONSIDERED, at its ninety-sixth session, amendments to the FSS Code proposed and circulated in accordance with article VIII(b)(i) of the Convention,

- 1) ADOPTS, in accordance with article VIII(b)(iv) of the Convention, amendments to the FSS Code the text of which is set out in the annex to the present resolution;
- 2) DETERMINES, in accordance with article VIII(b)(vi)(2)(bb) of the Convention, that the said amendments shall be deemed to have been accepted on 1 July 2019 unless, prior to the date, more than one third of the Contracting Governments to the Convention, or Contracting Governments the combined merchant fleets of which constitute not less than 50% of the gross tonnage of the world's merchant fleet, have notified the Secretary – General of their objections to the amendments.
- 3) INVITES Contracting Governments to the Convention to note that, in accordance with article VIII(b)(vii)(2) of the Convention, the amendments shall enter into force on 1 January 2020 upon their acceptance in accordance with paragraph 2 above;
- 4) REQUESTS the Secretary – General, for the purposes of Article VIII(b)(v) of the Convention, to transmit certified copies of the present resolution and the text of the amendments contained in the annex to all Contracting Governments to the Convention;
- 5 REQUESTS ALSO the Secretary-General to transmit copies of this resolution and its annex to Members of the Organization, which are not Contracting Governments to the Convention.



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ANNEX

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CHAPTER 8

AUTOMATIC SPRINKLER, FIRE DETECTION AND FIRE ALARM SYSTEMS

- 1) The text in existing paragraph 2.4.1 is replaced with the following:

"2.4.1 General

- 2.4.1.1 Any parts of the system which may be subjected to freezing temperatures in service shall be suitably protected against freezing.
- 2.4.1.2 Special attention shall be paid to the specification of water quality provided by the system manufacturer to prevent internal corrosion of sprinklers and clogging of blockage arising from products of corrosion of scale – forming minerals.

- 2) A new chapter 17 is added after existing chapter 16 as follows:

"CHAPTER 17

HELICOPTER FACILITY FOAM FIREFIGHTING APPLIANCES

1 Application

This chapter details the specifications for foam firefighting appliances for the protection of helidecks and helicopter landing areas as required by chapter II-2 of the Convention.

2 Definitions

- 2.1 *D-value* means the largest dimension of the helicopter used for assessment of the helideck when its rotors are turning. It establishes the required area of foam application.
- 2.2 Deck integrated foam nozzles are foam nozzles recessed into or edge mounted on the helideck.
- 2.3 *Foam-making branch pipes* are air-aspirating nozzles in tube shape for producing and discharging foam, usually in straight stream only.
- 2.4 *Helicopter landing area* is as defined in SOLAS regulation II-2/3.57.
- 2.5 *Helideck* is as defined in SOLAS II-2/3.26.
- 2.6 *Hose reel foam station* is a hose reel fitted with a foam-making branch pipe and non-collapsible hose, together with fixed foam proportioner and fixed foam concentrate tank, mounted on a common frame.



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- 2.7 *Monitor foam station* is a foam monitor, either self-inducing or together with separate fixed foam proportioner, and fixed foam concentrate tank, mounted on a common frame.
- 2.8 *Obstacle free sector* is the take-off and approach sector which totally encompasses the safe landing area and extends over a sector of at least 210°, within which only specified obstacles are permitted.
- 2.9 *Limited obstacle sector* is a 150° sector outside the take-off and approach sector that extends outward from a helideck where objects of limited height are permitted.

3 Engineering specifications for helidecks and helicopter landing areas

- 3.1 The system shall be capable of manual release and may be arranged for automatic release.
- 3.2 For helidecks the foam system shall contain at least two fixed foam monitors of deck integrated foam nozzles. IN addition, at least two hose reels fitted with a foam – making branch pipe and non-collapsible hose sufficient to reach any part of the helideck shall be provided. The minimum foam system discharge rate shall be determined by multiplying the D-value area by 6 //min/m². The minimum foam system discharge rate for deck integrated foam nozzle systems shall be determined by multiplying the overall helideck area by 6 //min/m². Each monitor shall be capable of supplying at least 50% of the minimum foam system discharge rate, but not less than 500//min. The minimum discharge rate of each hose reel shall be at least 400 // min. The quantity of foam concentrate shall be adequate to allow operation of all connected discharge devices for at least 5 min.
- 3.3 Where foam monitors are installed, the distance from the monitor to the farthest extremity of the protected area shall be not more than 75% of the monitor throw in still air conditions.
- 3.4 For helicopter landing areas, at least two portable foam applicators or two hose reel foam stations shall be provided, each capable of discharging a minimum foam solution discharge rate, in accordance with the following table.

Category	Helicopter overall length (D-value)	Minimum foam solution discharge rate (//min)
H1	up to but not including 15 m	250
H2	from 15 m up to but not including 24 m	500
H3	from 24 m up to but not including 35 m	800

The quantity of foam concentrate shall be adequate to allow operation of all connected discharge devices for at least 10 min. For tankers fitted with a deck foam system, the Administration may consider an alternative arrangement, taking into account the type of foam concentrate to be used.

- 3.5 Manual release stations capable of starting necessary pumps and opening required valves, including the fire main system, if used for water supply, shall be located at each monitor and hose reel. In addition, a central manual release station shall be provided at a protected location. The foam system shall be designed to discharge foam with nominal flow and at design pressure from any connected discharge devices within 30 s of activation.
- 3.6 Activation of any manual release station shall initiate the flow of foam solution to all connected hose reels, monitors, and deck integrated foam nozzles.



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- 3.7 The system and its components shall be designed to withstand ambient temperature changes, vibration, humidity, shock impact and corrosion normally encountered on the open deck and shall be manufactured and tested to the satisfaction of the Administration.
- 3.8 A minimum nozzle throw of at least 15 m shall be provided with all hose reels and monitors discharging foam simultaneously. The discharge pressure, flow rate and discharge pattern of deck integrated foam nozzles shall be to the satisfaction of the Administration, based on tests that demonstrate the nozzle's capability to extinguish fires involving the largest size helicopter for which the helideck is designed.
- 3.9 Monitors, foam-making branch pipes, deck integrated foam nozzles and couplings shall be constructed of brass, bronze or stainless steel. Piping, fittings and related components, except gaskets, shall be designed to withstand exposure to temperatures up to 925°C.
- 3.10 The foam concentrate shall be demonstrated effective for extinguishing aviation fuel spill fires and shall conform to performance standards not inferior to those acceptable to the Organization. Where the foam storage tank is on the exposed deck, freeze protected foam concentrates shall be used, if appropriate, for the area of operation.
- 3.11 Any foam system equipment installed within the take-off and approach obstacle-free sector shall not exceed a height of 0.25 m. Any foam system equipment installed in the limited obstacle sector shall not exceed the height permitted for objects in this area.
- 3.12 All manual release stations monitor foam stations, hose reel foam stations, hose reels and monitors shall be provided with a means of access that does not require travel across the helideck or helicopter landing area.
- 3.13 Oscillating monitors, if used, shall be pre-set to discharge foam in a spray pattern and have a means of disengaging the oscillating mechanism to allow rapid conversion to manual operation.
- 3.14 If a foam monitor with flow rate up to 1000 l/min is installed, it shall be equipped with an air-aspirating nozzle. If a deck integrated nozzle system is installed, then the additionally installed hose reel shall be equipped with an air-aspirating handling nozzle (foam branch pipes). Use of non-air aspirating foam nozzles (on both monitors and the additional hose reel) is permitted only where foam monitors with a flow rate above 1000 l/min are installed. If only portable foam applicators or hose reel stations are provided, these shall be equipped with an air-aspirating handline nozzle (foam branch pipes).

